

Rajalakshmi Engineering College

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NeoColab_REC_CS23231_DATA STRUCTURES

REC_DS using C_Week 6_COD_Question 1

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

John and Mary are collaborating on a project that involves data analysis. They each have a set of age data, one sorted in ascending order and the other in descending order. However, their analysis requires the data to be in ascending order.

Write a program to help them merge the two sets of age data into a single sorted array in ascending order using merge sort.

Input Format

The first line of input consists of an integer N, representing the number of age values in each dataset.

The second line consists of N space-separated integers, representing the ages of participants in John's dataset (in ascending order).

The third line consists of N space-separated integers, representing the ages of participants in Mary's dataset (in descending order).

Output Format

The output prints a single line containing space-separated integers, which represents the merged dataset of ages sorted in ascending order.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

1 3 5 7 9

10 8 6 4 2

Output: 1 2 3 4 5 6 7 8 9 10

Answer

```
#include <stdio.h>
```

```
void merge(int arr[], int left[], int right[], int left_size, int right_size) {  
    int i = 0, j = 0, k = 0;  
    while(i < left_size && j < right_size) {  
        if(left[i] <= right[j]) arr[k++] = left[i++];  
        else arr[k++] = right[j++];  
    }  
    while(i < left_size) arr[k++] = left[i++];  
    while(j < right_size) arr[k++] = right[j++];  
}
```

```
void mergeSort(int arr[], int size) {  
    if(size < 2) return;  
    int mid = size / 2;  
    int left[mid], right[size - mid];  
  
    for(int i = 0; i < mid; i++) left[i] = arr[i];  
    for(int i = mid; i < size; i++) right[i - mid] = arr[i];  
  
    mergeSort(left, mid);  
    mergeSort(right, size - mid);  
}
```

```
    merge(arr, left, right, mid, size - mid);
}

int main() {
    int n, m;
    scanf("%d", &n);
    int arr1[n], arr2[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr1[i]);
    }
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr2[i]);
    }
    int merged[n + n];
    mergeSort(arr1, n);
    mergeSort(arr2, n);
    merge(merged, arr1, arr2, n, n);
    for (int i = 0; i < n + n; i++) {
        printf("%d ", merged[i]);
    }
    return 0;
}
```

Status : Correct

Marks : 10/10